

Student: Jonatan Rabjerg
Date: 1/15/24

Instructor: Andreas Basse-O'Connor
Course: Calculus Beta

Assignment: MyLab Math - Uge 3

Evaluate the integral using integration by parts.

$$\int x \sin \left(\frac{1}{3}x \right) dx$$

To evaluate $\int x \sin \left(\frac{1}{3}x \right) dx$, use integration by parts which has the general form $\int u dv = uv - \int v du$.

With $u = x$ and $dv = \sin \left(\frac{1}{3}x \right) dx$, then $du = dx$ and $v = -3 \cos \left(\frac{1}{3}x \right)$.

The completed integration is shown.

$$\begin{aligned} \int x \sin \left(\frac{1}{3}x \right) dx &= -3x \cos \left(\frac{1}{3}x \right) - \int -3 \cos \left(\frac{1}{3}x \right) dx \\ &= -3x \cos \left(\frac{1}{3}x \right) + 9 \sin \left(\frac{1}{3}x \right) + C \end{aligned}$$